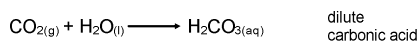


Acid Rain

Normal rain is slightly acidic (pH = 5.6) due to natural levels of CO₂, NO₂, and SO₂ in the air.

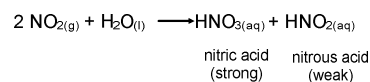


Normal occurrences of CO₂:

- respiration
- forest fires
- volcanic eruptions

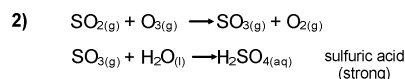
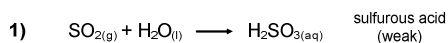
Nitrogen oxides occur naturally from:

- lightning strikes
- plant decay



Sulfur oxides occur naturally from:

- volcanic eruptions

**Acid Precipitation**

Any form of natural precipitation (rain, snow, etc.) that has an unusually high acidity (pH < 5.6)

Acid Deposition

- acid forming pollutants
- includes acid precipitation
- also includes dust and other dry particulate matter

Table 1 Effect of Water pH on Aquatic Life in Lakes and Rivers

pH	Effect
6.0	• Crustaceans, insects, and some plankton species begin to disappear.
5.0	• Changes occur in the plankton community. • Less desirable species of mosses and plankton appear. • Some fish populations die out. Smallmouth bass, walleye, brook trout, and salmon are the least tolerant of acidity. • Most fish eggs cannot hatch.
<5.0	• There are no fish. • The lake bottom is covered with undecayed material. • The surrounding shoreline may be dominated by mosses. • Terrestrial animals that depend on aquatic ecosystems are affected.

Table taken from *General Chemistry: 12 College Preparation*, p. 205

Unit 4 Review

p487 #1-17, 19-24, 26, 27, 37, 29

p492 #1-21, 24, 25

